



BASEBALL AND STATISTICS

A LOVE AFFAIR

Course Leaders



Steve A

Bats: Left	Throws: Right
Height: 6-0	Weight: 185
Born: 12-9-30 Laurens, So. Car.	
Robert Sidney	
"Hurricane"	
HAZLE	
Outfielder (1)	
11 - 6	31 - 8
12 - 25	32 - 26
13 - 14	33 - 6
14 - 30	34 - 31
15 - 7	35 - 8
16 - 28	36 - 33
21 - 32	41 - 26
22 - 7	42 - 9
23 - 32	43 - 29
24 - 13	44 - 7
25 - 7	45 - 14
26 - 40	46 - 13
51 - 7	52 - 27
53 - 15	54 - 32
55 - 7	56 - 12
61 - 8	62 - 14
63 - 29	64 - 9
65 - 35	66 - 1

Tom N



Tom R



Course Agenda

- Oct 13th
 - History and Context of Baseball Statistics
 - Basic Metrics (Batting, Pitching and Fielding)
 - Examples with Minnesota Twins Players
 - Modern Influences for Advanced Analytics
 - Sabermetrics, Moneyball
- Oct 20th
 - Advanced Analytics
 - Can Advanced Analytics be Used to Make Comparisons of Players from Different Eras
- October 27th
 - In Game Metrics and Decision Making
 - Data Analytics and Overall Team Strategy
- November 3rd
 - Predictive Modeling
 - Ethics and Fairplay
 - Use of AI



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SESSION 1 HISTORY AND EVOLUTION OF BASEBALL STATISTICS



Baseball Statistics

- Statistics and Data Analytics Have Always Been a Part of Baseball and Have Evolved with the Game
 - Deadball Era, The Yankee Dynasty, The Designated Hitter, The Steroid Era, Today's Game which is Driven by Data Analytics
- Historically Were Presented in Box Scores for Individual Games
- Season and Career Statistics Have Been Used to Compare Player Performance
- Basic Statistics Have Included
 - **Batting**-Batting Average, Homeruns, Runs Batted In
 - **Pitching**-Wins and Losses, Earned Run Average
 - **Fielding**-Chances, Errors, Fielding Percentage

Boxscores and MLB Averages (Sunday Newspaper)

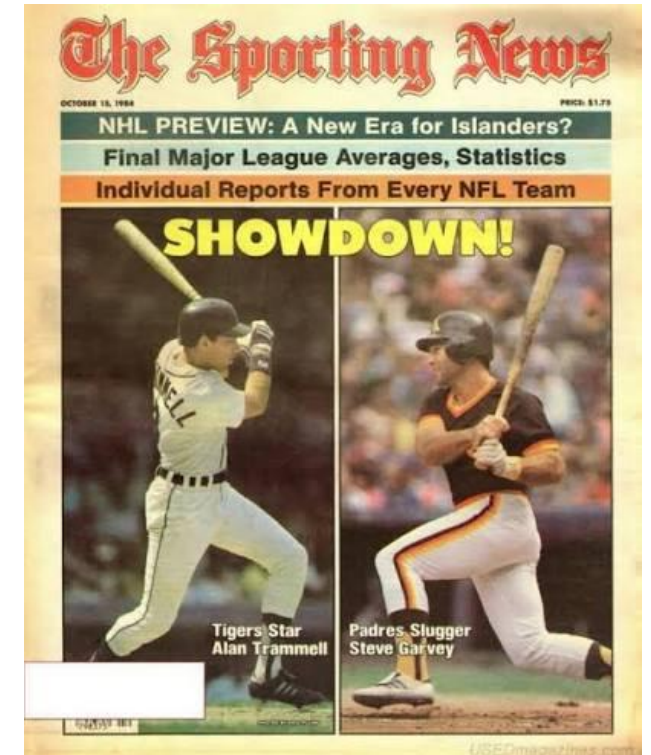
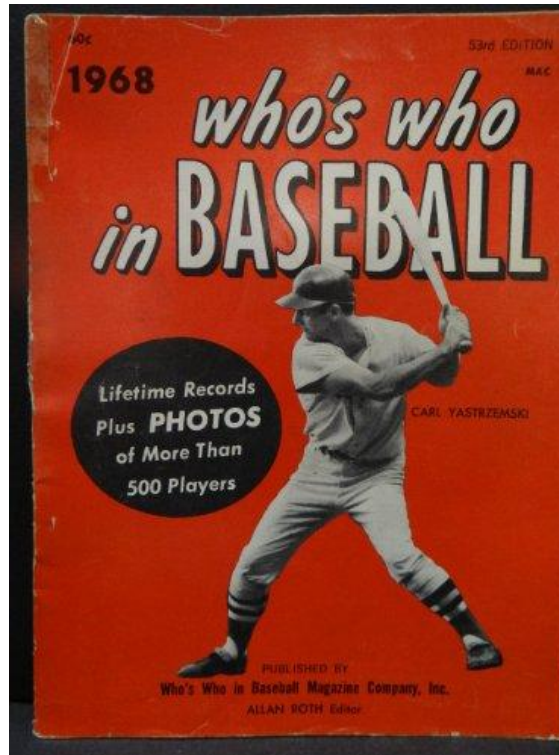
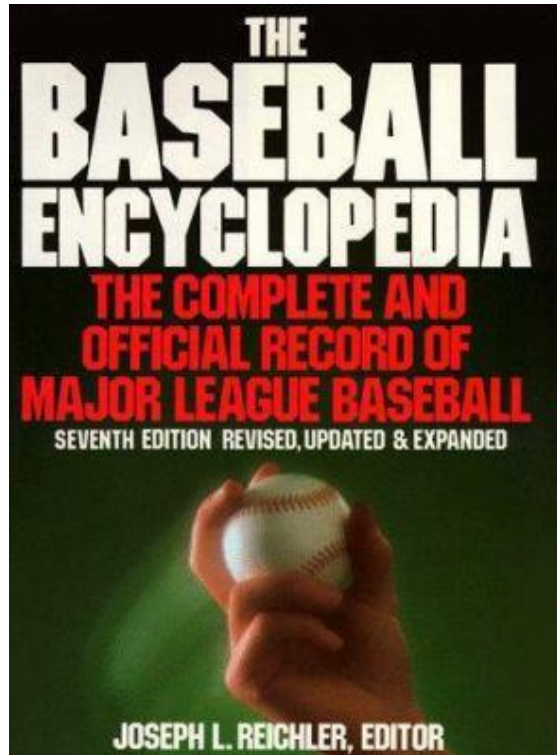
GAME 7 BOX SCORE										
TWINS 1, BRAVES 0 (10)										
Atlanta	AB	R	H	BI	BB	SO	Avg			
LoSmith dh	4	0	2	0	1	1	.231			
Pendleton 3b	5	0	1	0	0	0	.367			
Gant cf	4	0	0	0	0	2	.267			
Justice rf	3	0	1	0	1	1	.259			
Bream lf	4	0	0	0	0	0	.125			
Hunter lf	4	0	1	0	0	1	.190			
Olson c	4	0	0	0	0	1	.222			
Lemke 2b	4	0	1	0	0	1	.417			
Bellard ss	2	0	1	0	0	1	.375			
c-Blauser ph-ss	1	0	0	0	0	0	.167			
Totals	35	0	7	0	2	8				
www.baseball-almanac.com										
Minnesota	AB	R	H	BI	BB	SO	Avg			
Gladden lf	5	1	3	0	0	1	.233			
Knoblauch 2b	4	0	1	0	0	0	.306			
Puckett cf	2	0	0	0	3	1	.250			
Hrbek lf	3	0	0	0	1	0	.115			
CDavis dh	4	0	1	0	0	1	.222			
2-JBrown pr-ph	0	0	0	0	0	0	.000			
d-Larkin ph	1	0	1	1	0	0	.500			
Harper c	4	0	2	0	0	0	.381			
Mack rf	4	0	1	0	0	0	.130			
Paglanulo 3b	3	0	0	0	1	0	.273			
Gagne ss	2	0	0	0	0	1	.167			
a-Bush ph	1	0	1	0	0	0	.250			
1-Newman pr-ss	0	0	0	0	0	0	.500			
b-Sorrento ph	1	0	0	0	0	1	.000			
Leius ss	0	0	0	0	0	0	.357			
Totals	34	1	10	1	5	5				
Atlanta	000	000	000	0	0	7	.0			
Minnesota	000	000	000	1	1	10	.0			
One out when winning run scored.										
a-singled for Gagne in the 8th. b-struck out for Newman in the 9th. c-fouled out for Bellard in the 10th. d-singled for Brown in the 10th.										
1-ran for Bush in the 8th. 2-ran for Davis in the 9th.										
LOB—Atlanta 8, Minnesota 12. 2B—Pendleton (3), Hunter (1), Gladden 2 (2). RBIs—Larkin (1). S—Bellard, Knoblauch, GIDP—Bream, CDavis, Mack.										
Runners left in scoring position—Atlanta 5 (Gant 2, Bream, Olson 2); Minnesota 4 (Puckett, Hrbek, Paglanulo, Sorrento).										
Runners moved up—Bream, Knoblauch.										
DP—Atlanta 3 (Bream, Bellard and Bream), (Lemke), (Lemke, Bellard and Bream); Minnesota 1 (Hrbek, Harper and Hrbek).										
Atlanta	IP	R	H	ER	BB	SO	NP	ERA		
Smoltz	7	1	3	6	0	0	1	4.104	1.26	
Stanton	2	3	2	0	0	1	0	16	0.00	
Pena L.O-1	1	1	3	2	1	1	3	1	12	3.38
Minnesota	IP	R	H	ER	BB	SO	NP	ERA		
Morris W, 2-0	10	7	0	0	2	8	122	1.17		
Stanton pitched to 2 batters in the 9th.										
Inherited runners scored—Stanton 2-0, Pena 2-0.										
IBB—off Stanton (Puckett) 1, off Pena (Puckett) 1, off Pena (Hrbek) 1, off Pena (Paglanulo) 1, off Morris (Justice) 1.										
HBP—by Smoltz (Hrbek). PB—Harper.										
Umpires—Home, Denkinger; First, Wendelstedt; Second, Coble; Third, Tata; Left, Reed; Right, Montague.										
T—3:23. A—55,118.										

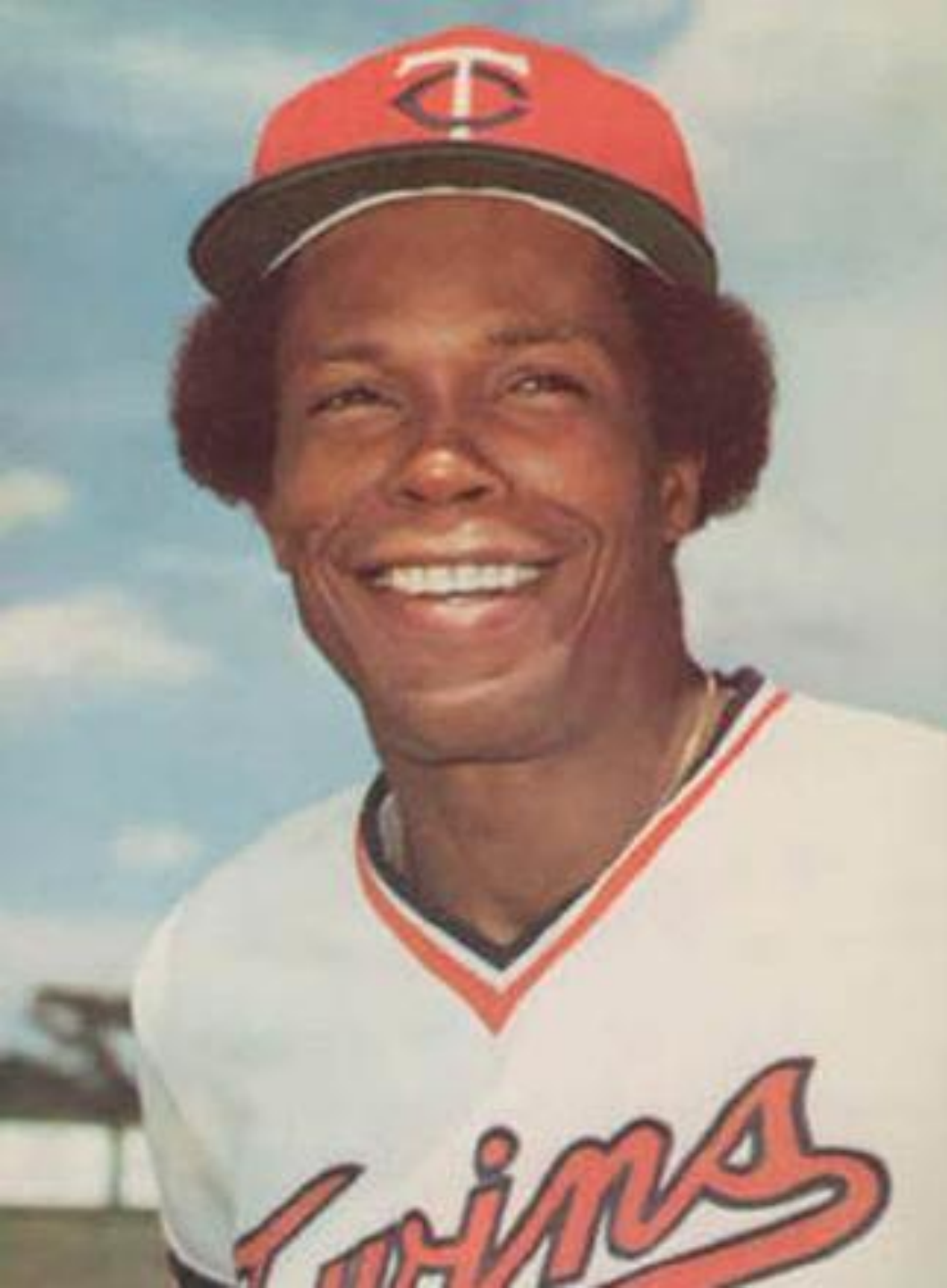
major league averages

National League						American League							
Team Batting						Team Batting							
	AB	R	H	HR	RBI	Pct		AB	R	H	HR	RBI	Pct
St. Louis	1769	248	485	28	227	.274	Kansas City	1710	243	497	29	237	.291
Pittsburgh	1716	229	464	41	217	.270	Cleveland	1666	254	466	41	237	.280
Cincinnati	1721	189	465	21	174	.270	Boston	1719	227	473	38	209	.275
Atlanta	1667	225	439	45	205	.264	Toronto	1753	219	481	22	207	.275
San Diego	1700	239	445	26	215	.262	Chicago	1696	231	462	38	220	.272
Los Angeles	1771	217	463	36	207	.262	Milwaukee	1686	237	447	60	220	.265
New York	1744	213	454	36	199	.260	Detroit	1640	221	431	36	206	.263
Montreal	1569	188	405	38	180	.258	California	1764	223	462	43	209	.262
Philadelphia	1730	199	408	26	186	.250	Baltimore	1660	230	433	53	219	.261
San Francisco	1765	205	439	38	190	.249	Seattle	1904	206	482	37	191	.253
Houston	1709	186	414	24	177	.242	New York	1668	214	416	36	199	.249
Chicago	1744	202	422	26	188	.242	Texas	1527	157	380	30	149	.249
							Minnesota	1825	195	445	51	183	.244
							Oakland	1776	241	410	46	224	.231

Individual Batting						Individual Batting							
75 or more at bats						75 or more at bats							
	AB	R	H	HR	RBI	Pct		AB	R	H	HR	RBI	Pct
RuJones SD	169	41	59	8	32	.349	Harrah Cle	190	44	75	10	29	.395
JThompson Pgh	174	34	59	13	37	.339	Bonnell Tor	129	26	49	3	21	.380
Green STL	77	8	26	0	14	.338	McBride Cle	85	8	31	0	13	.365
Kulper SF	80	9	27	0	9	.338	Cooper Mil	192	32	67	9	36	.359
Bailor NY	95	12	32	0	13	.337	McRae KC	185	27	65	8	45	.351
Ramsey STL	84	8	28	0	5	.333	Lowensn Bal	95	25	32	10	25	.337
Wohlford SF	109	18	36	2	12	.330	Yastzmski Bos	141	18	47	8	29	.333
Knight Htn	200	29	65	3	26	.325	WWilson KC	09	14	36	1	12	.330
Stearns NY	170	28	55	3	14	.324	Grubb Tex	85	5	28	2	11	.329
Driessen Cin	181	23	58	7	26	.320	Piniella NY	82	16	27	2	16	.329
Guerrero LA	192	29	61	10	35	.318	Hrbek Min	168	28	55	11	35	.327
Wiggins SD	101	23	32	1	3	.317	Paciorek Chi	165	19	54	3	22	.327
Oliver Mil	171	25	54	5	29	.316	Herdon Det	187	31	61	5	30	.326
JRay Pgh	195	30	61	3	26	.313	Hebner Det	86	14	28	0	8	.326
Francona Mtl	84	9	26	0	3	.310	EMurray Bal	123	16	40	6	23	
Moreland Chi	194	22	60	9	39	.309		3	2	5			
LoSmith STL	205	46	63	5	31	.307	Upshaw Tor	171	23	55	3	23	.322
Bliffner Cin	88	14	27	2	10	.307	Iorg Tor	119	14	38	0	12	.319
Cncepcion Cin	194	21	5	1	24	.304	Garcia Tor	202	26	64	1	15	.317
Sax LA	204	33	62	0	14	.304	Thornton Cle	181	40	57	16	51	.315
Wallach Mtl	159	20	48	6	25	.302	RWshgtn Min	143	12	45	2	10	.315
Dawson Mtl	190	37	57	9	28	.300	Carew Cal	159	27	50	0	12	.314
Baker LA	177	17	53	8	31	.299	Bochte Sea	179	14	56	2	22	.313
Cedeno Cin	157	15	47	1	20	.299	Otis KC	177	33	55	3	33	.311
Horner Atl	154	34	46	10	28	.299	Eisenreich Min	88	9	27	2	8	.307
Wilson NY	213	32	63	2	22	.296	Cabell Det	184	19	54	2	22	.304
Dernier Phil	149	33	44	2	9	.295							
Lester SD	177	30	43	6	23	.294							

Historical Sources of Baseball Statistics





BATTING STATISTICS

Harmon Killebrew 1965



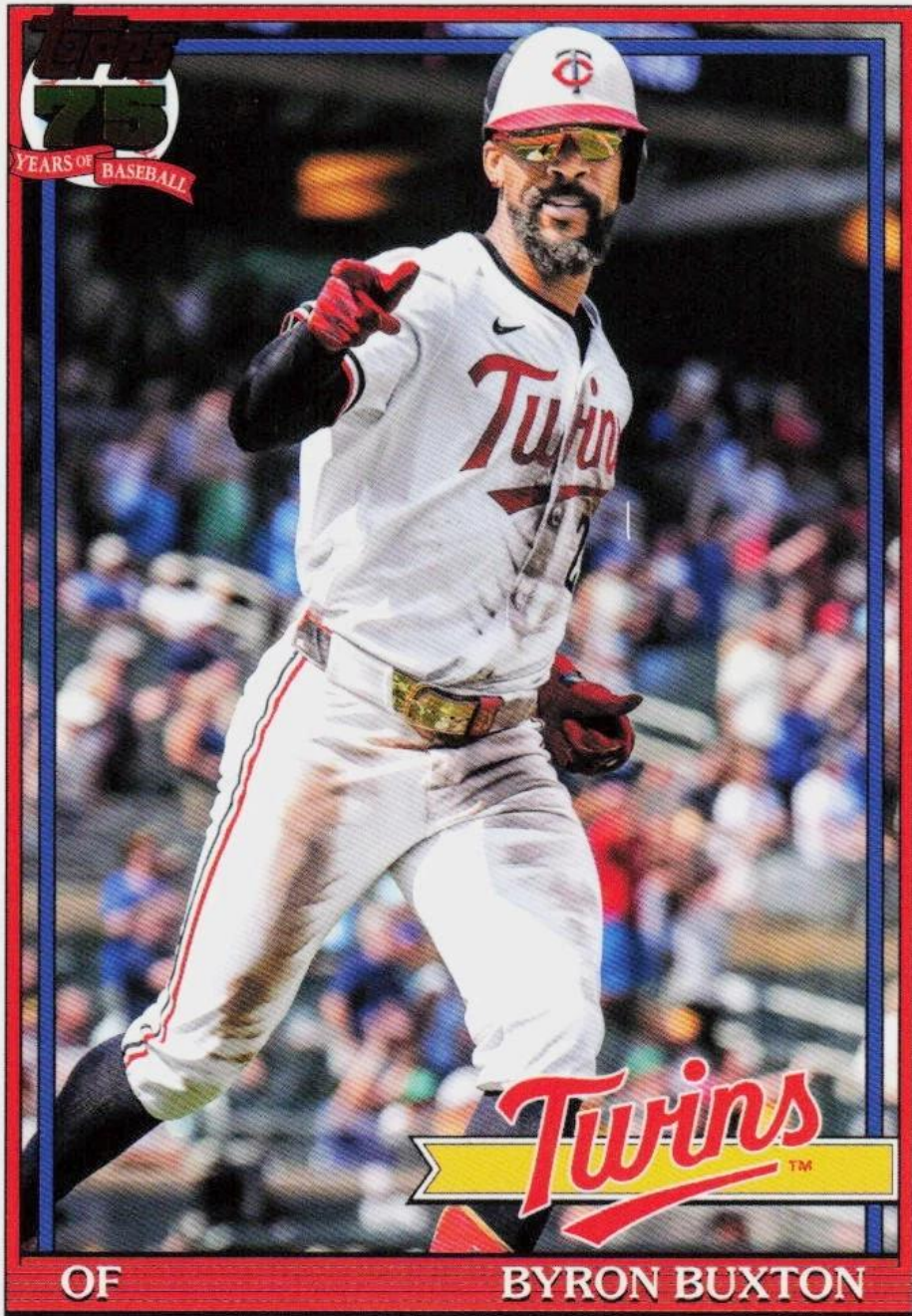
TOPPS 400 **HARMON KILLEBREW**
IN 1964, HARMON LED
THE A.L. IN HOME RUNS
FOR THE THIRD STRAIGHT YEAR.

HEIGHT: 5'11" WEIGHT: 213 BATS: RIGHT THROWS: RIGHT
BORN: JUNE 29, 1936 HOME: ONTARIO, ORE.

COMPLETE MAJOR LEAGUE BATTING RECORD

YEAR	TEAM	LEA.	G	AB	R	H	2B	3B	HR	RBI	AVG.
1954	Washington	A. L.	9	13	1	4	1	0	0	3	.308
1955	Washington	A. L.	38	80	12	16	1	0	4	7	.200
1956	Washington	A. L.	44	99	10	22	2	0	5	13	.222
1957	Washington	A. L.	9	31	4	9	2	0	2	5	.290
1958	Washington	A. L.	13	31	2	6	0	0	0	2	.194
1959	Washington	A. L.	153	546	98	132	20	2	42	105	.242
1960	Washington	A. L.	124	442	84	122	19	1	31	80	.276
1961	Minnesota	A. L.	150	541	94	156	20	7	46	122	.288
1962	Minnesota	A. L.	155	552	85	134	21	1	48	126	.243
1963	Minnesota	A. L.	142	515	88	133	18	0	45	96	.258
1964	Minnesota	A. L.	158	577	95	156	11	1	49	111	.270
Major League Totals 11 Yrs.			995	3427	573	890	115	12	272	670	.260

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Byron Buxton 2025

Topps

T91-92

HT: 6'2" WT: 190 BATS: RIGHT THROWS: RIGHT DRFTD: TWINS #1-JUNE, 2012
ACQ: VIA DRAFT BORN: 12-18-93, BAXLEY, GA. HOME: BAXLEY, GA.

MINNESOTA TWINS®

BYRON BUXTON • OF

COMPLETE MAJOR LEAGUE BATTING RECORD

YR	CLUB	G	AB	R	H	2B	3B	HR	RBI	SB	SLG	BB	SO	AVG
2015	TWINS	46	129	16	27	7	1	2	6	2	.326	6	44	.209
2016	TWINS	92	298	44	67	19	6	10	38	10	.430	23	118	.225
2017	TWINS	140	462	69	117	14	6	16	51	29	.413	38	150	.253
2018	TWINS	28	90	8	14	4	0	0	4	5	.200	3	28	.156
2019	TWINS	87	271	48	71	30	4	10	46	14	.513	19	68	.262
2020	TWINS	39	130	19	33	3	0	13	27	2	.577	2	36	.254
2021	TWINS	61	235	50	72	23	0	19	32	9	.647	13	62	.306
2022	TWINS	92	340	61	76	13	3	28	51	6	.526	34	116	.224
2023	TWINS	85	304	49	63	17	1	17	42	9	.438	35	109	.207
2024	TWINS	102	355	62	99	27	3	18	56	7	.524	20	99	.279
2025	TWINS	126	488	97	129	21	7	35	83	24	.551	41	148	.264

MAJOR LEA. TOTALS	898	3102	523	768	178	31	168	436	117	.487	234	978	.248
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Byron can carry his team with hot streaks. "When this guy turns it on, he really turns it on," skipper Rocco Baldelli says. "He's doing it on a nightly basis, and it just keeps getting better and better."

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MLB PLAYERS

Batting Average (Hits ÷ At Bats)

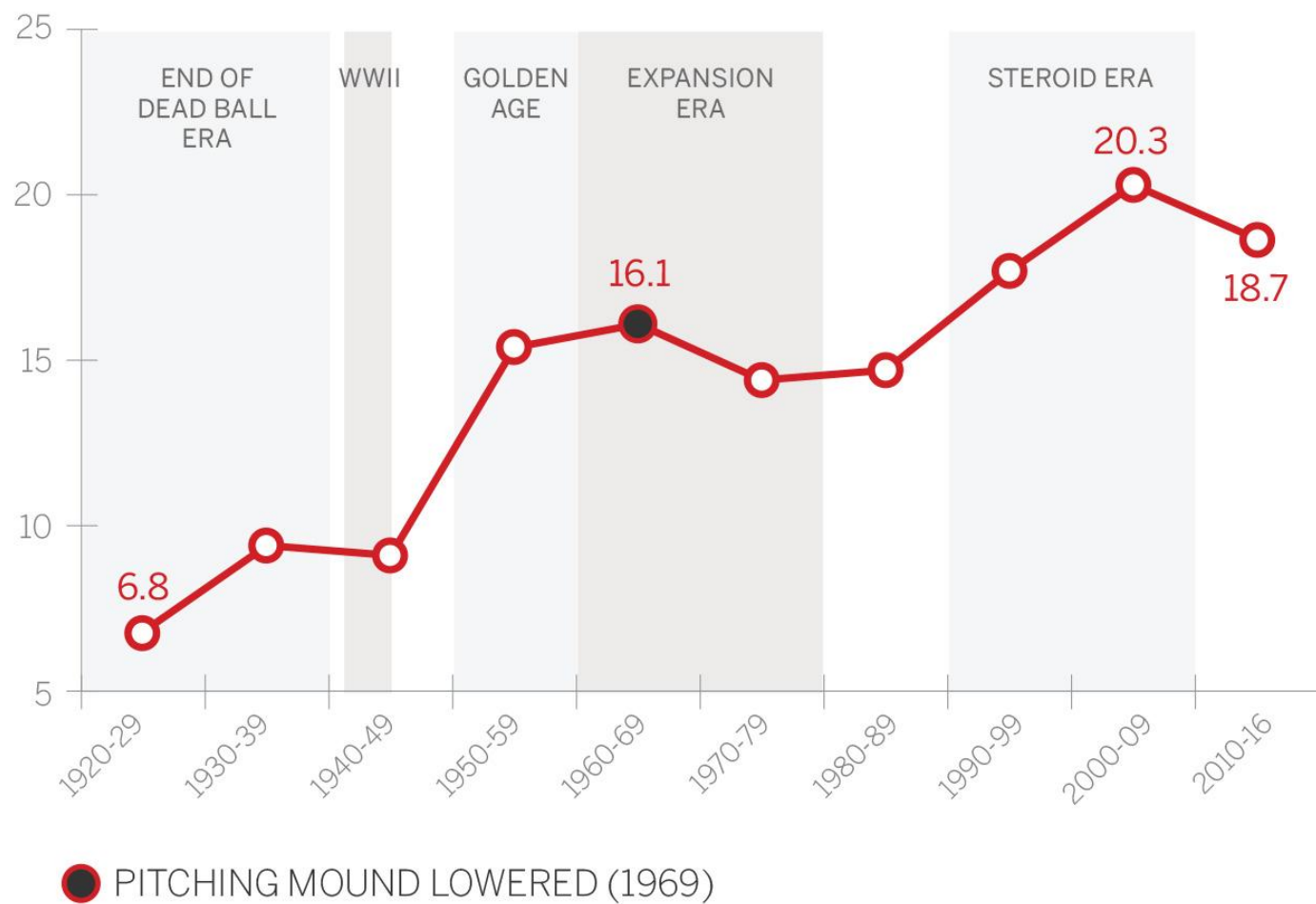
Era	League AVG	What Was Happening
1930	.296	Live ball era, tiny ballparks, no relief specialists
1950	.266	Integration, better pitching depth
1968	.237	Year of the Pitcher (MLB lowered the mound after this)
1980	.265	Power renaissance beginning
2000	.270	Steroid era peak, juiced balls, small strike zones
2010	.257	Post-steroid correction
2019	.252	Last “normal” season before shifts and spider tack
2023	.248	Shift ban boosted average slightly
2025	.245	Velocity arms race, expanding strike zones

Batting Average	What It Means	2025 Examples
.330+	MVP-caliber, historically elite	Aaron Judge (.331)
.300-.329	All-Star, batting title contender	Bo Bichette (.311), Trea Turner (.304)
.280-.299	Above average, consistent producer	Freddie Freeman (.295), Bobby Witt Jr. (.295)
.250-.279	League average	Where most everyday starters land
.230-.249	Below average	Playable if the player has power or walks
Under .230	Struggling	A problem for anyone without elite defense

Limitations of Basic Hitting Statistics

- All hits are treated equally
 - Extra-base hits are the same as bloop singles
- Doesn't account for on-base skills
 - Walks and HBP reflect a player's ability to get on base but are not accounted for in Batting Average
- Ignores context
 - League hitting rates, ballpark dimensions, quality of opposing pitcher
- Doesn't account for historical shifts
 - Good batting average depends upon the era

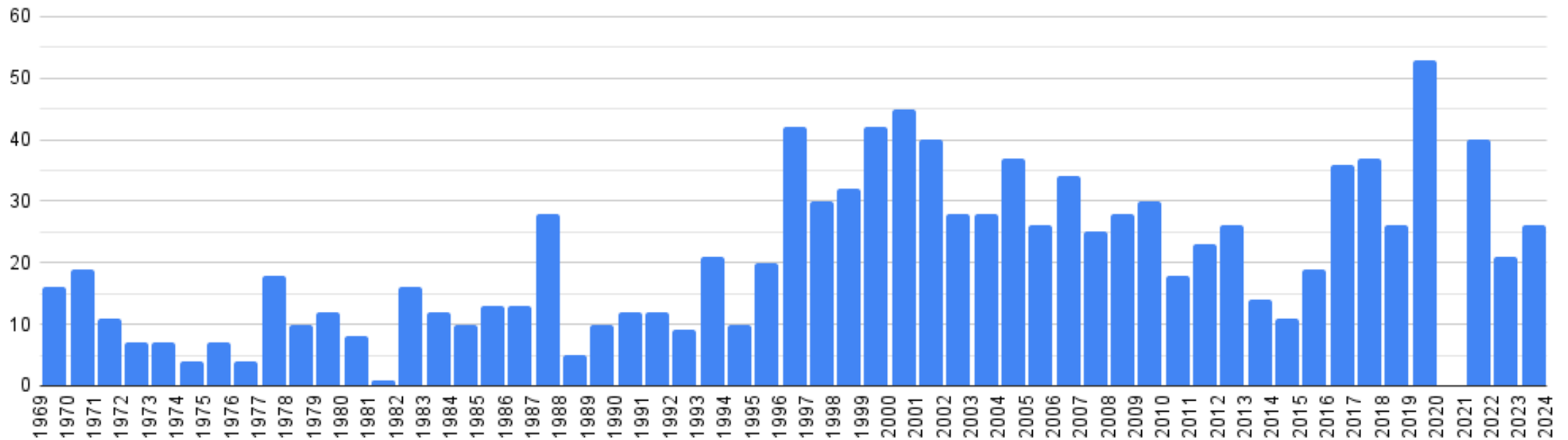
HR's/Game by Era



Increase in HRs (Players with 30+ HRs)

Players with 30+ Home Runs in a Season

MLB 1969-2023



Slugging Percentage (SLG)

- SLG is calculated by Total Bases/At Bats
 - Singles+Doublesx2+Triplesx3+Home Runsx4/At Bats
- Examples
 - Hank Aaron(1959)
 - 131 singles+46x2 doubles+7x3 triples+39x4 homeruns
 - 400 total bases/629 at bats = .636
 - Rod Carew(1977)
 - SLG of ,570 with 38 doubles, 18 triples and 14 homeruns
 - Not just dependent on HRs

Rating	SLG
Excellent	.550 or Above
Great	.500
Above Average	,450
Average	.420
Below Average	.400
Poor	.390
Awful	.380 or Below

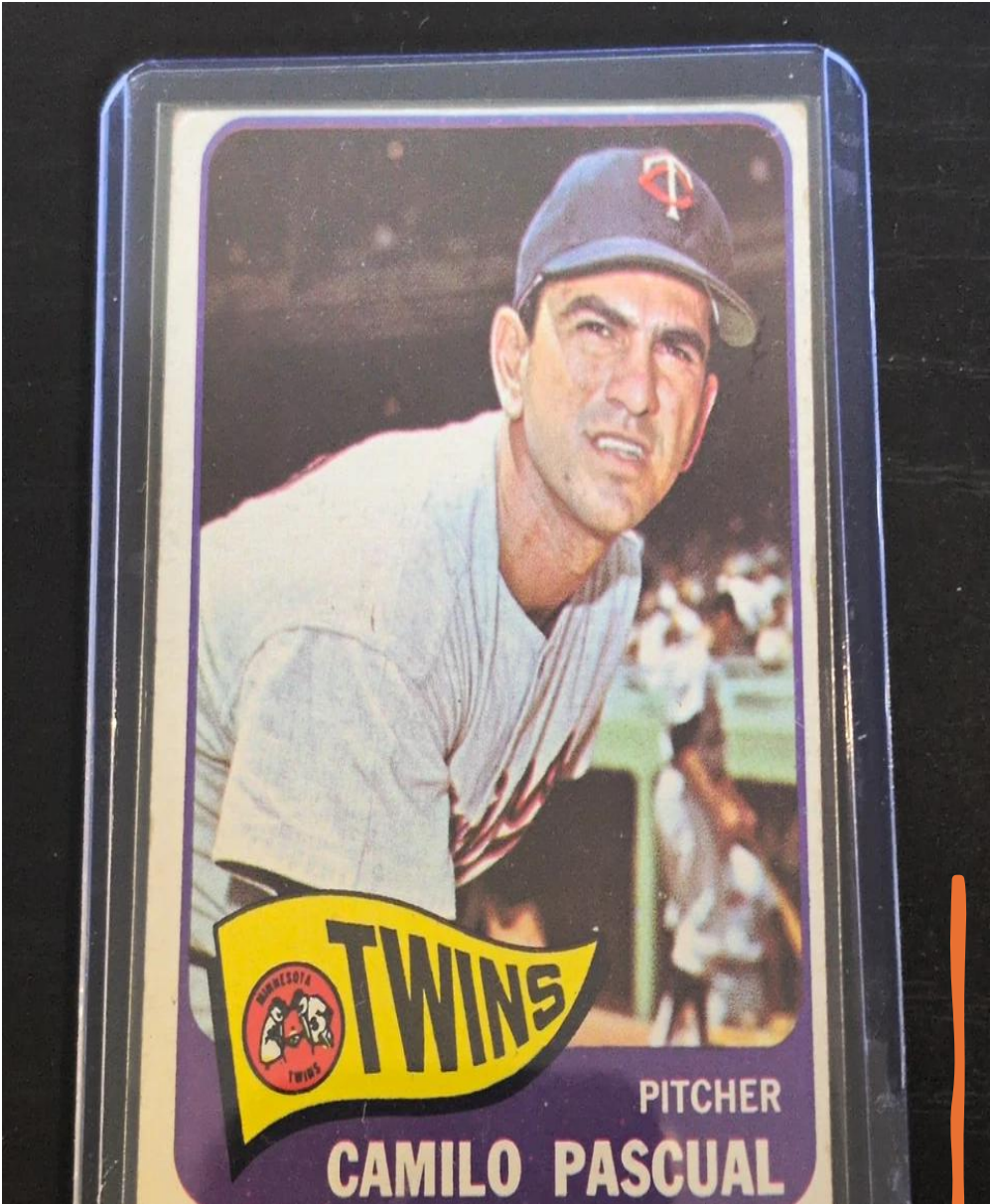
Complementary Statistics for BA





PITCHING STATISTICS

Camilo Pascual 1965



TOPPS 255

CAMILO PASCUAL

CAMILO LED THE AMERICAN LEAGUE WITH 6 SHUTOUTS IN 1959.

HEIGHT: 5'11" WEIGHT: 185 THROWS: RIGHT BATS: RIGHT
BORN: JANUARY 20, 1934 HOME: BLOOMINGTON, MINN.

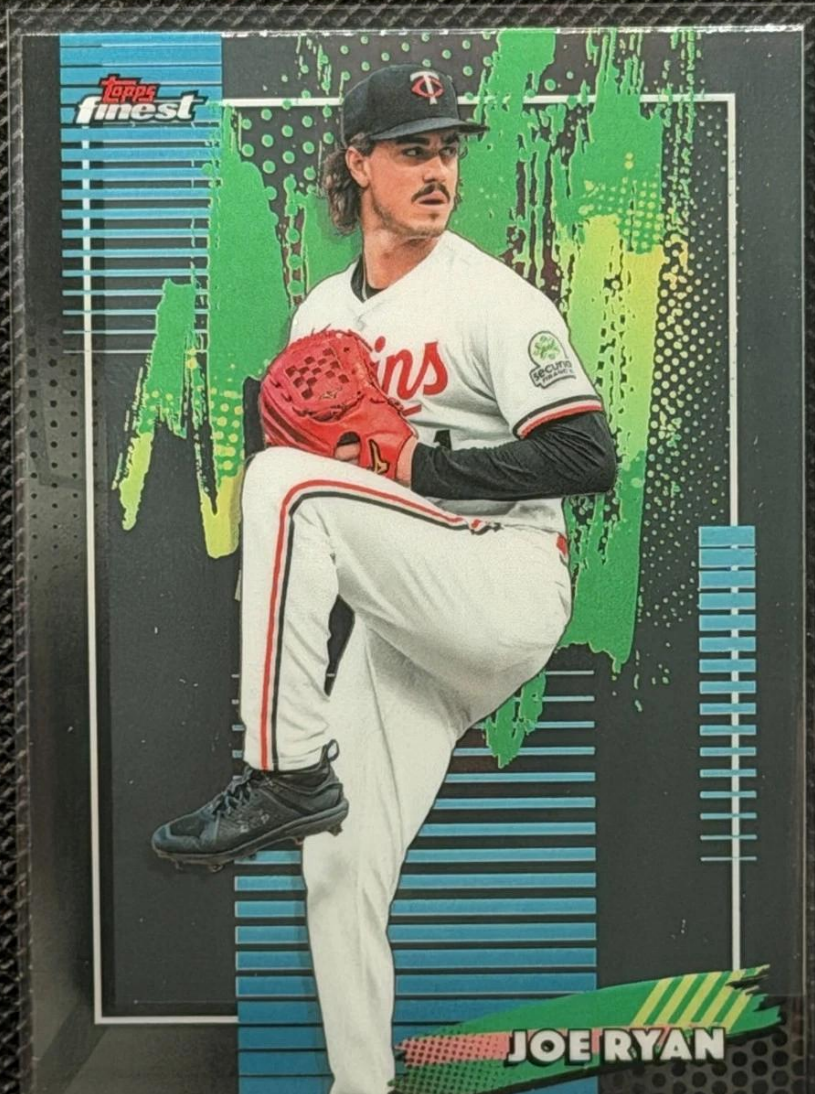
In 1964, Camilo finished second among A.L. pitchers with 14 complete games.

COMPLETE MAJOR LEAGUE PITCHING RECORD

YEAR	TEAM	LEA.	G	IP	W	L	PCT	SO	BB	ERA	
1954	Washington	A. L.	48	119	4	7	.364	60	61	4.24	
1955	Washington	A. L.	43	129	2	12	.143	82	70	6.14	
1956	Washington	A. L.	39	189	6	18	.250	162	89	5.86	
1957	Washington	A. L.	29	176	8	17	.320	113	76	4.09	
1958	Washington	A. L.	31	177	8	12	.400	146	60	3.15	
1959	Washington	A. L.	32	239	17	10	.630	185	69	2.64	
1960	Washington	A. L.	26	152	12	8	.600	143	53	3.02	
1961	Minnesota	A. L.	35	252	15	16	.484	221	100	3.46	
1962	Minnesota	A. L.	34	258	20	11	.645	206	59	3.31	
1963	Minnesota	A. L.	31	248	21	9	.700	202	81	2.47	
1964	Minnesota	A. L.	36	267	15	12	.556	213	98	3.30	
Major League Totals			11 Yrs.	384	2206	128	132	.492	1733	816	3.62

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Joe Ryan 2026



COMMON

JOE RYAN

MINNESOTA TWINS® - P

HT: 6'2" WT: 205 THROWS: RIGHT BATS: RIGHT
BORN: 6-5-96, SAN FRANCISCO, CALIFORNIA
DRAFTED: RAYS #7-JUNE, 2018

FINEST 2025 MAJOR LEAGUE PERFORMANCE

Joe sparked on April 27, spinning seven shutout innings to lead the Twins to a 5-0 victory over the visiting Angels. Allowing just four hits and one walk, Ryan fanned 11 while throwing 67 of 98 pitches for strikes.

MAJOR LEAGUE PITCHING RECORD

	G	IP	W	L	CG	SHO	BB	SO	WHIP	ERA	WAR
2025	31	171	13	10	0	0	39	194	1.04	3.42	4.5
CAREER	115	641.1	46	36	1	1	148	719	1.06	3.79	11.0

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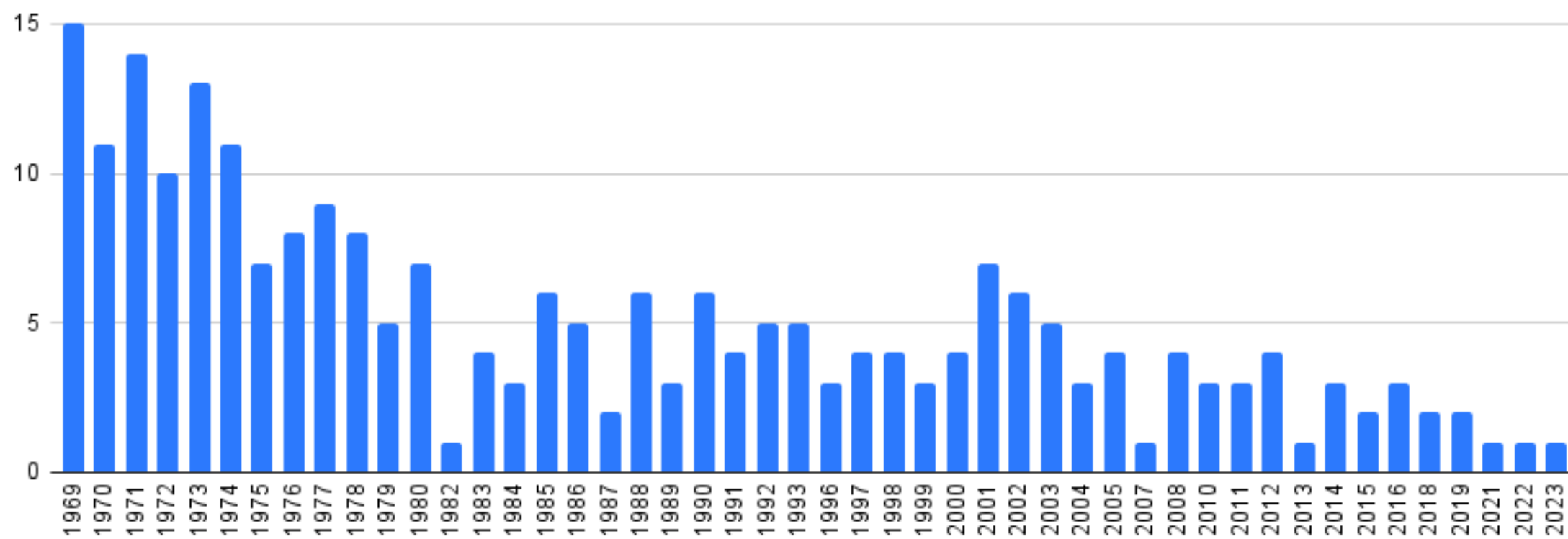
MLB

PLAYERS

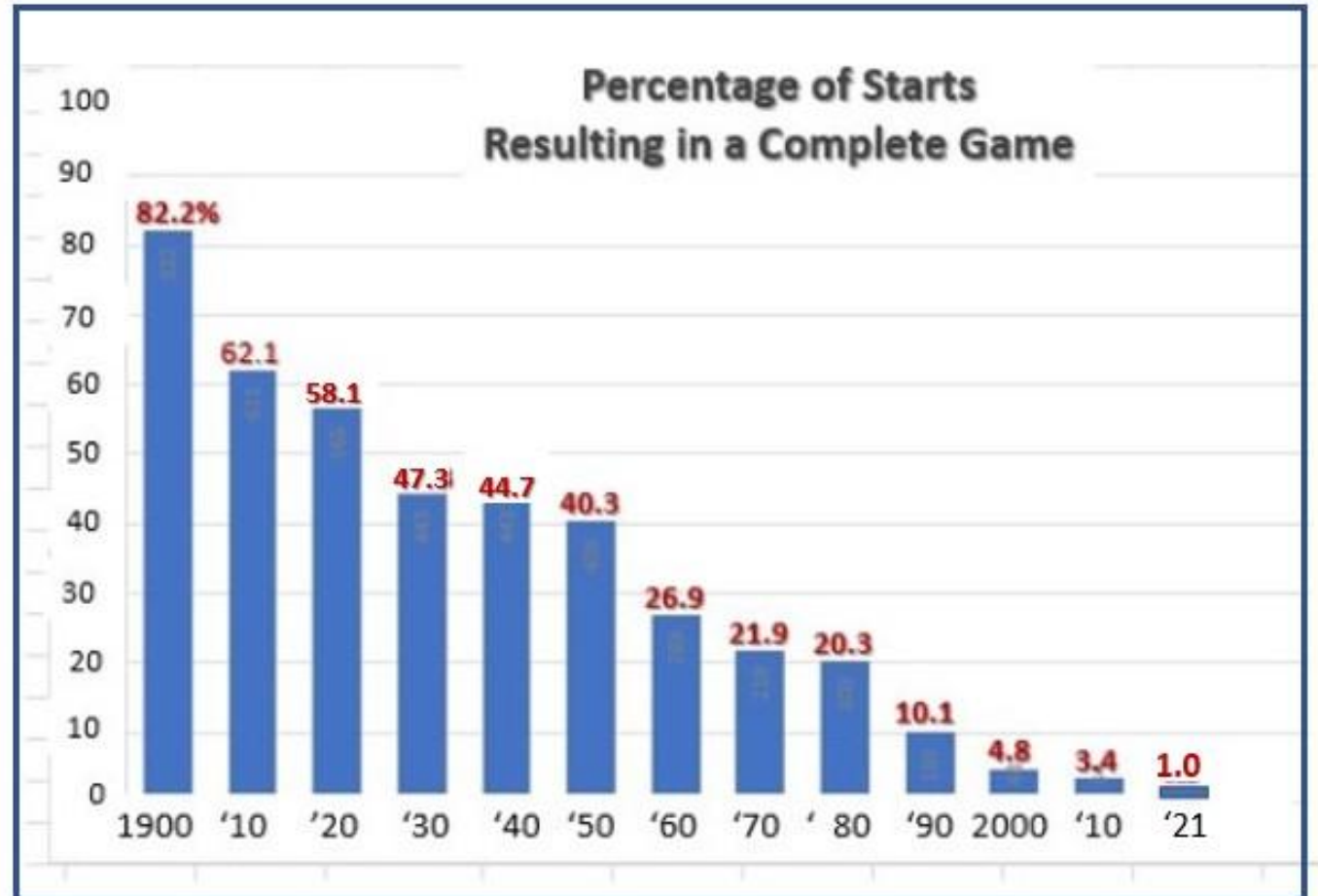
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Number of 20 game winners

MLB 1969-2023



Decrease in Number of Complete Games

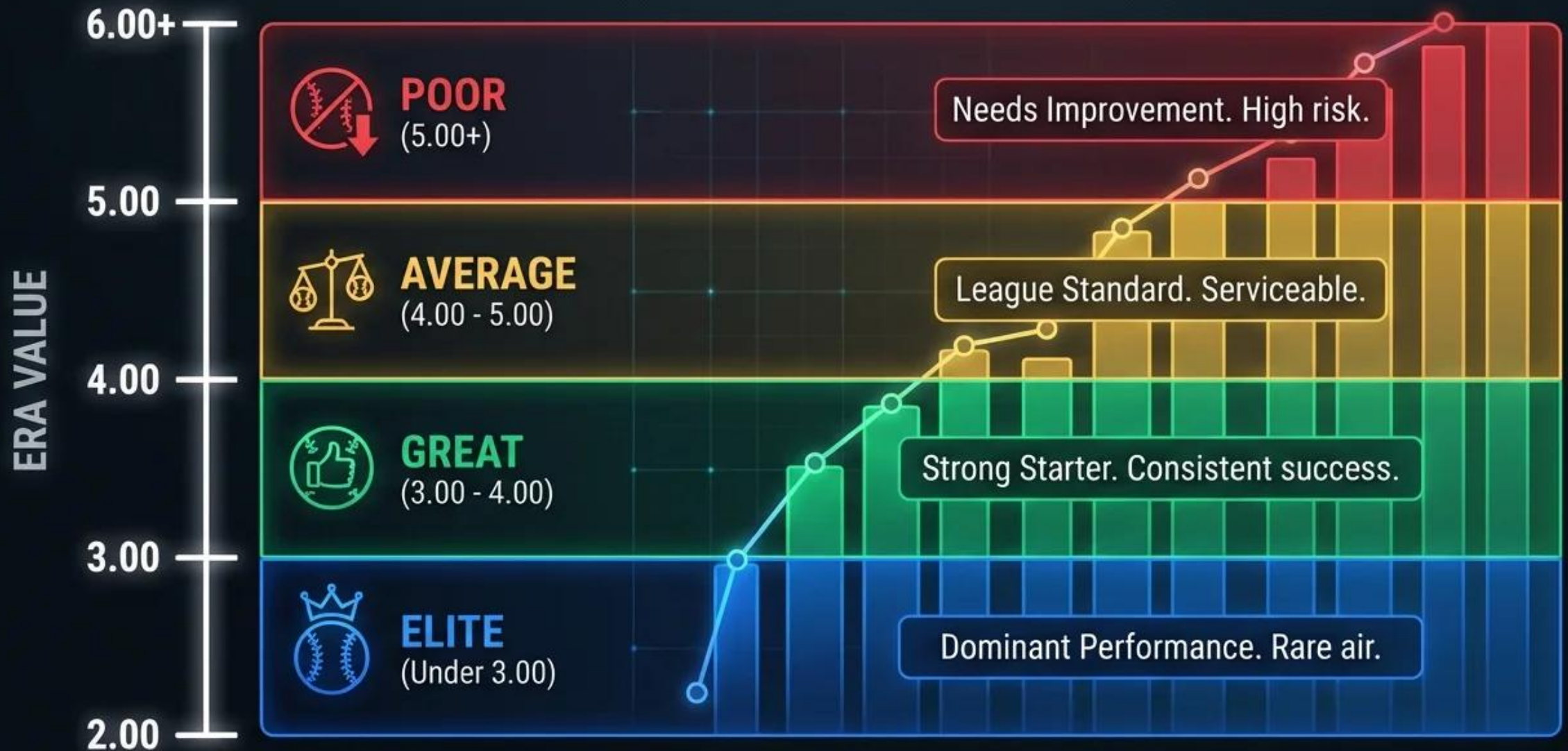




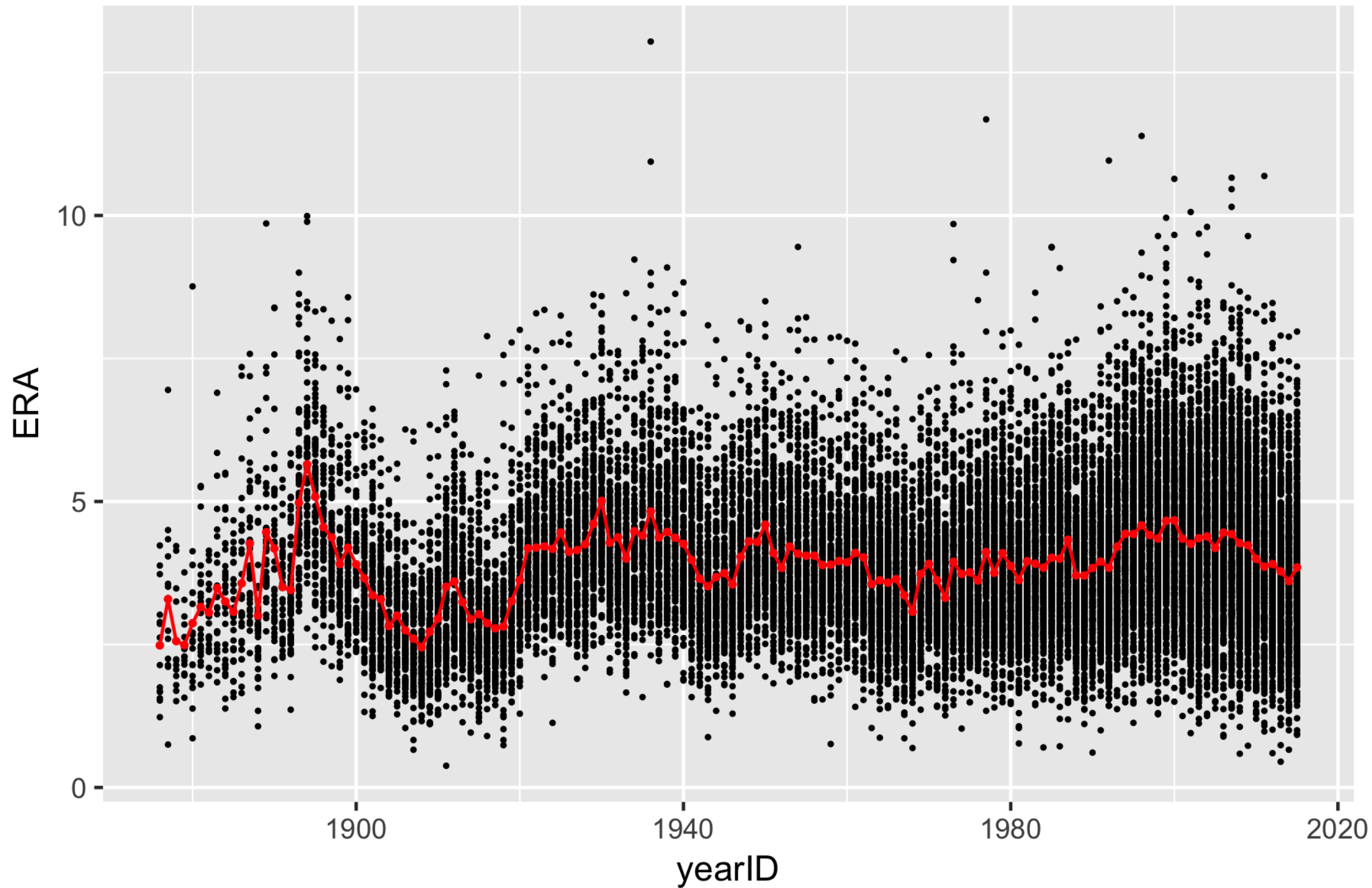
$$\text{ERA} = 9 \times \left(\frac{\text{Earned Runs}}{\text{Innings Pitched}} \right)$$



ERA (EARNED RUN AVERAGE) BENCHMARKS FOR PITCHERS



Note: Lower ERA indicates better pitching performance. Benchmarks are approximate and can vary by league and era.



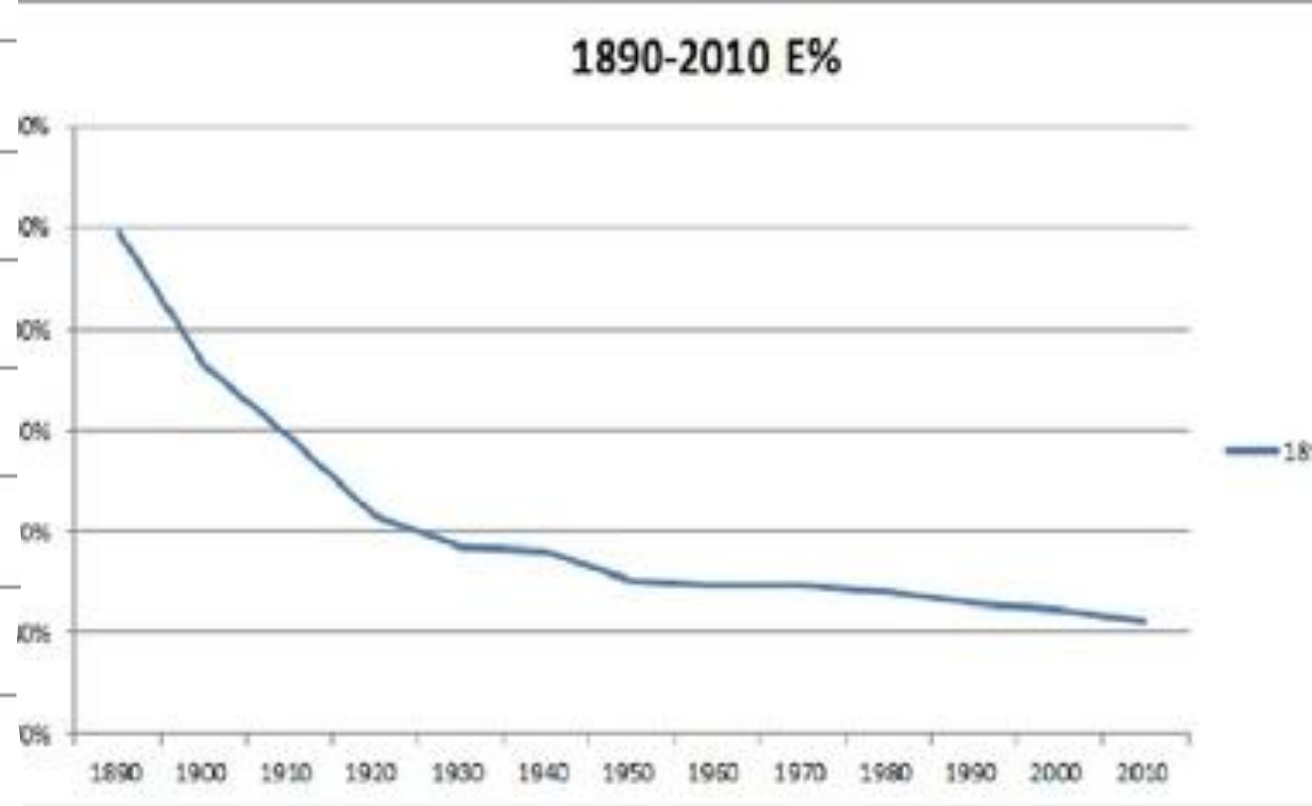
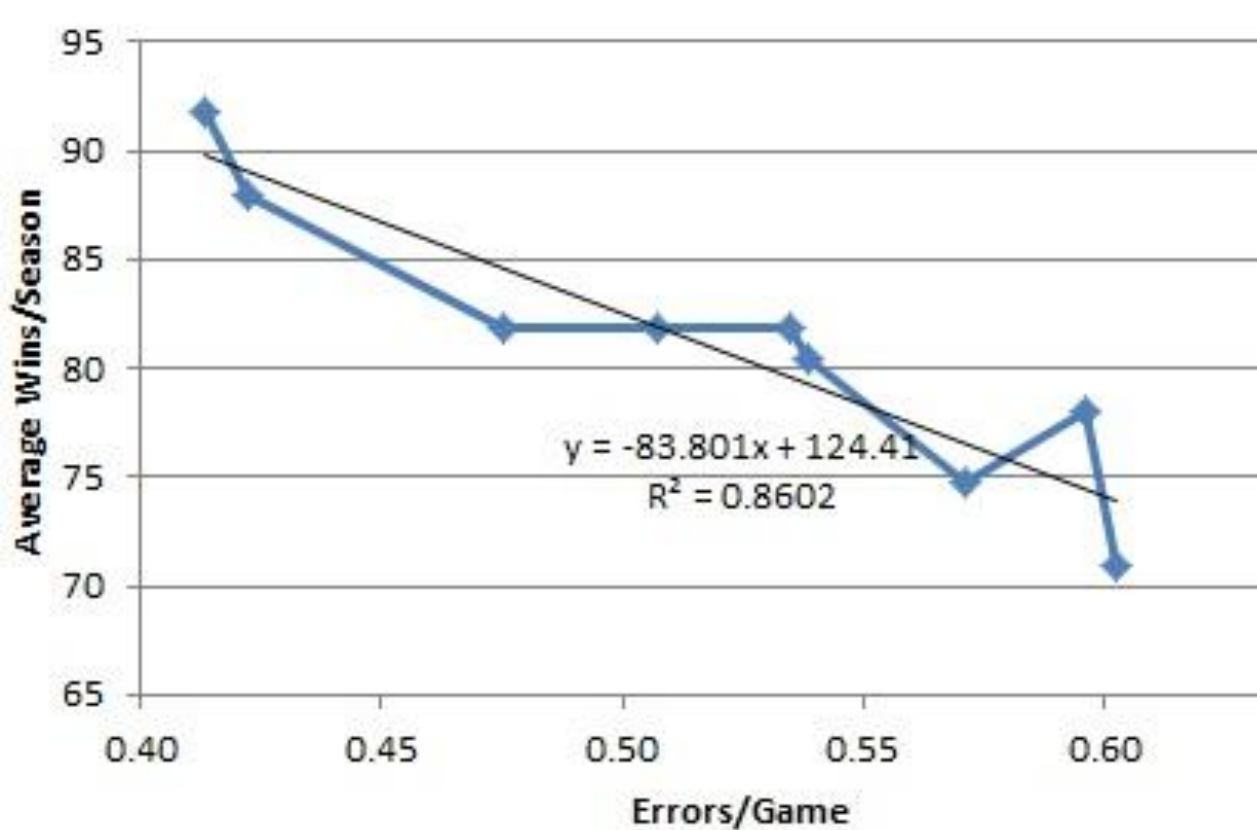
Current Pitching Trends

- Starters usually go 5-6 innings
 - Rarely pitch a 3rd time through the batting order
 - Focus on Quality Starts
 - 5+ innings, with 3 or fewer runs
- Increased use of the bullpen
 - Number of relief pitchers on each team
 - Focus on velocity
 - Metrics
 - Holds and Saves





FIELDING METRICS



Fielding Data

- Fewer Errors per Game
 - Better equipment
- Fewer Errors Correlates with Winning
- Recognized by Gold Glove Award
 - Often not the Player with the Best Stats



STATISTICS AND TWINS BASEBALL

CLASS PARTICIPATION TIME



BEST BATTING
AVERAGE

SEASON AND CAREER

Top Batting Average

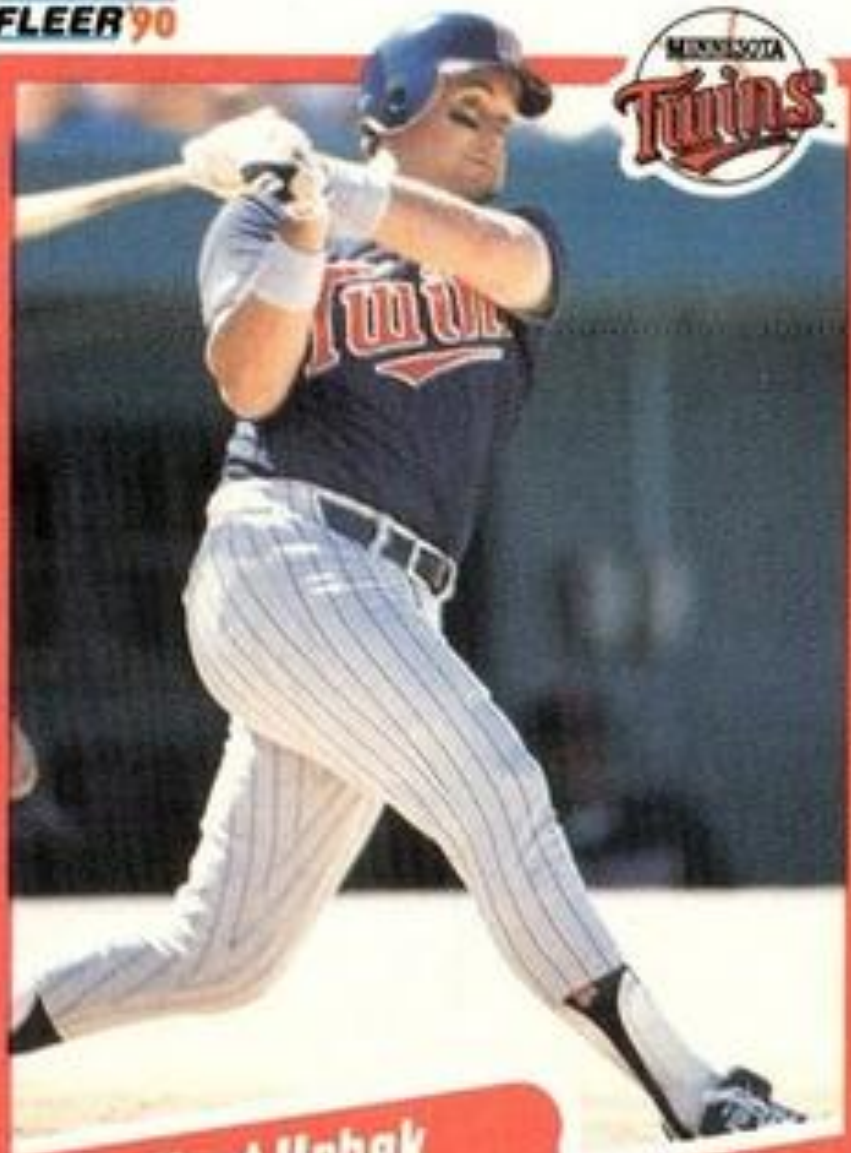
Season

1. Rod Carew	.388
2. Joe Mauer	.365
3. Rod Carew	.364
4. Rod Carew	.359
5. Kirby Puckett	.356

Career

1. Rod Carew	.334
2. Kirby Puckett	.318
3. Lyman Bostock	.318
4. Luis Arraez	.314
5. Paul Molitor	.312

FLEER '90



Kent Hrbek

FIRST BASE

HOMERUNS

SEASON AND CAREER

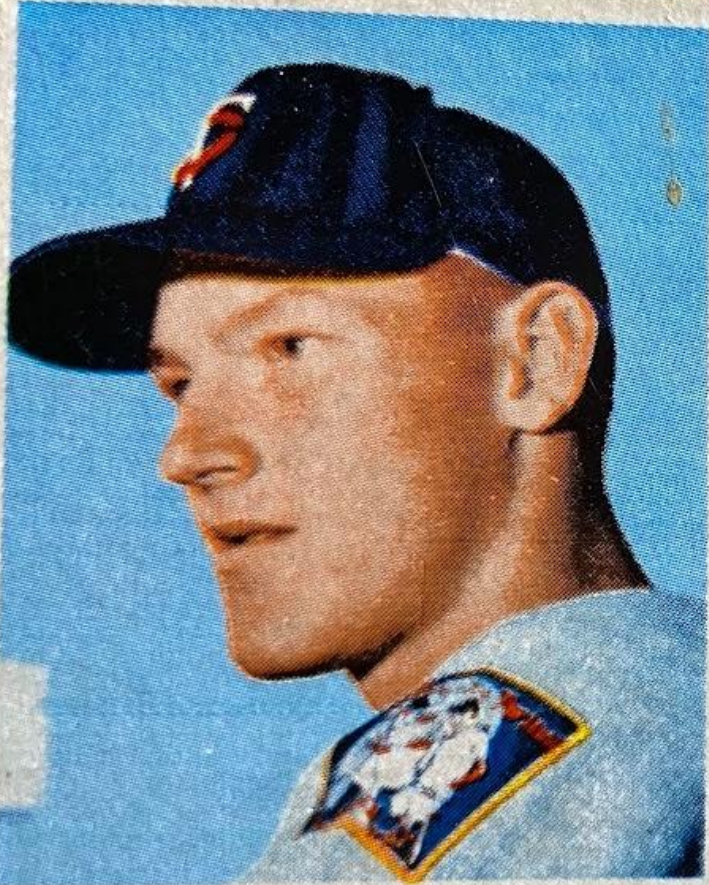
Home Runs

Season

1. Harmon Killebrew	49
2. Harmon Killebrew	48
3. Harmon Killebrew	46
4. Harmon Killebrew	45
5. Harmon Killebrew	44

Career

1. Harmon Killebrew	549
2. Kent Hrbek	293
3. Bob Allison	256
4. Justin Morneau	221
5. Tony Oliva	220



MINNE

Ht. 6'4"; Wt.
Born Nov. 7,
Full-fledged s
for a great s
his pro career
had a fine y
and led the P
innings pitch
16, 1959, he s
Nashville. Jim
and 1960 s
switched with



MAJOR LEAGUE PITCHING

	Games	IP	Won	Lost	Pcts.	Hits	Runs
1962	39	269	18	14	.563	243	106
LIFE	91	525	28	38	.424	486	259

PITCHING WINS

SEASON AND CAREER

Pitching Wins

Season

1. Kim Kaat	25
2. Jim Perry	24
3. Frank Viola	24

Career

1. Jim Kaat	190
2. Bert Blyleven	149
3. Brad Radke	148
4. Camillo Pascual	145
5. Jim Perry	128



ERA

SEASON AND CAREER

Lowest Earned Run Average

Season

Career

Best Fielding Center Fielder

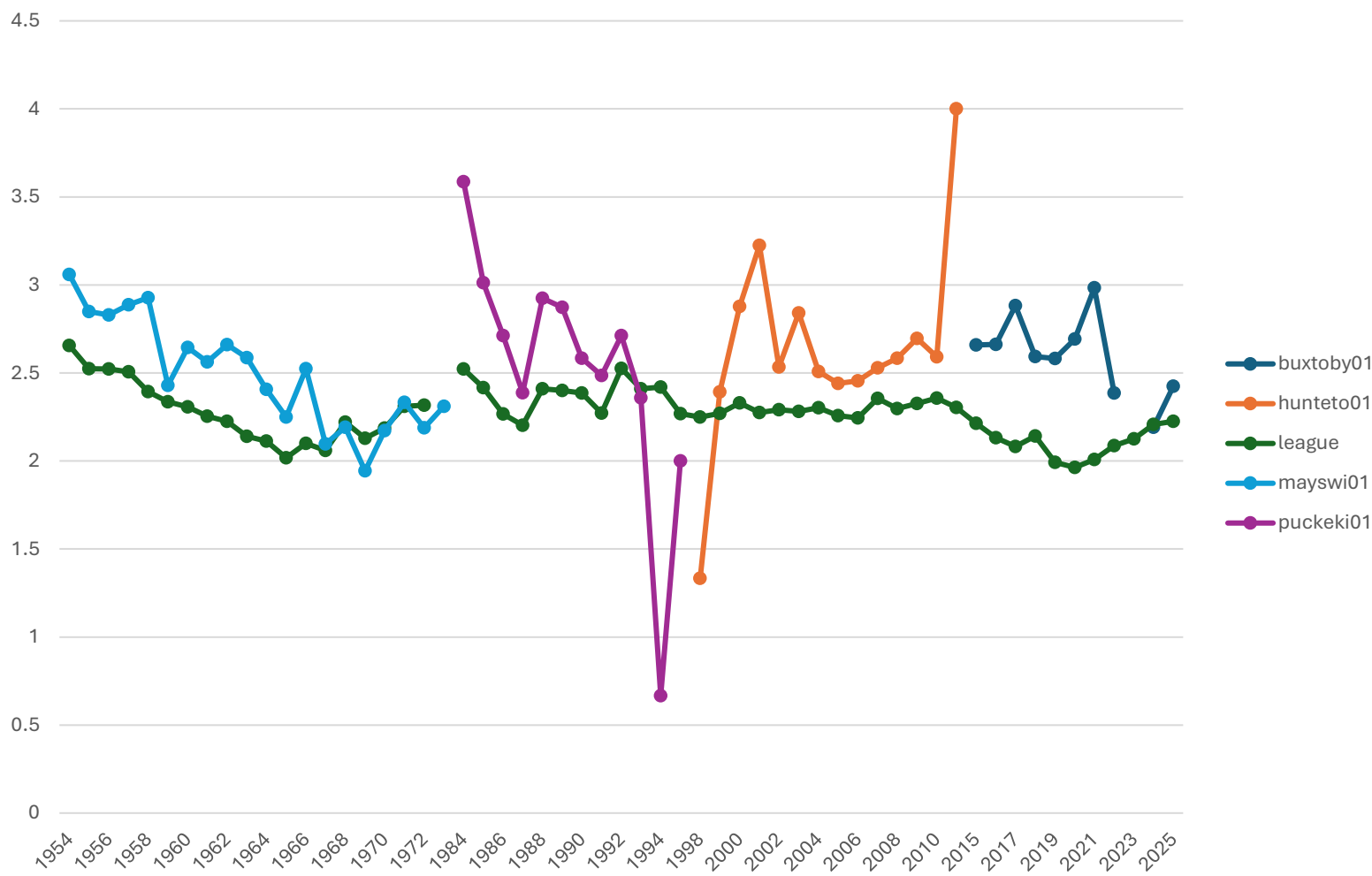
- Byron Buxton
- Tori Hunter
- Kirby Puckett



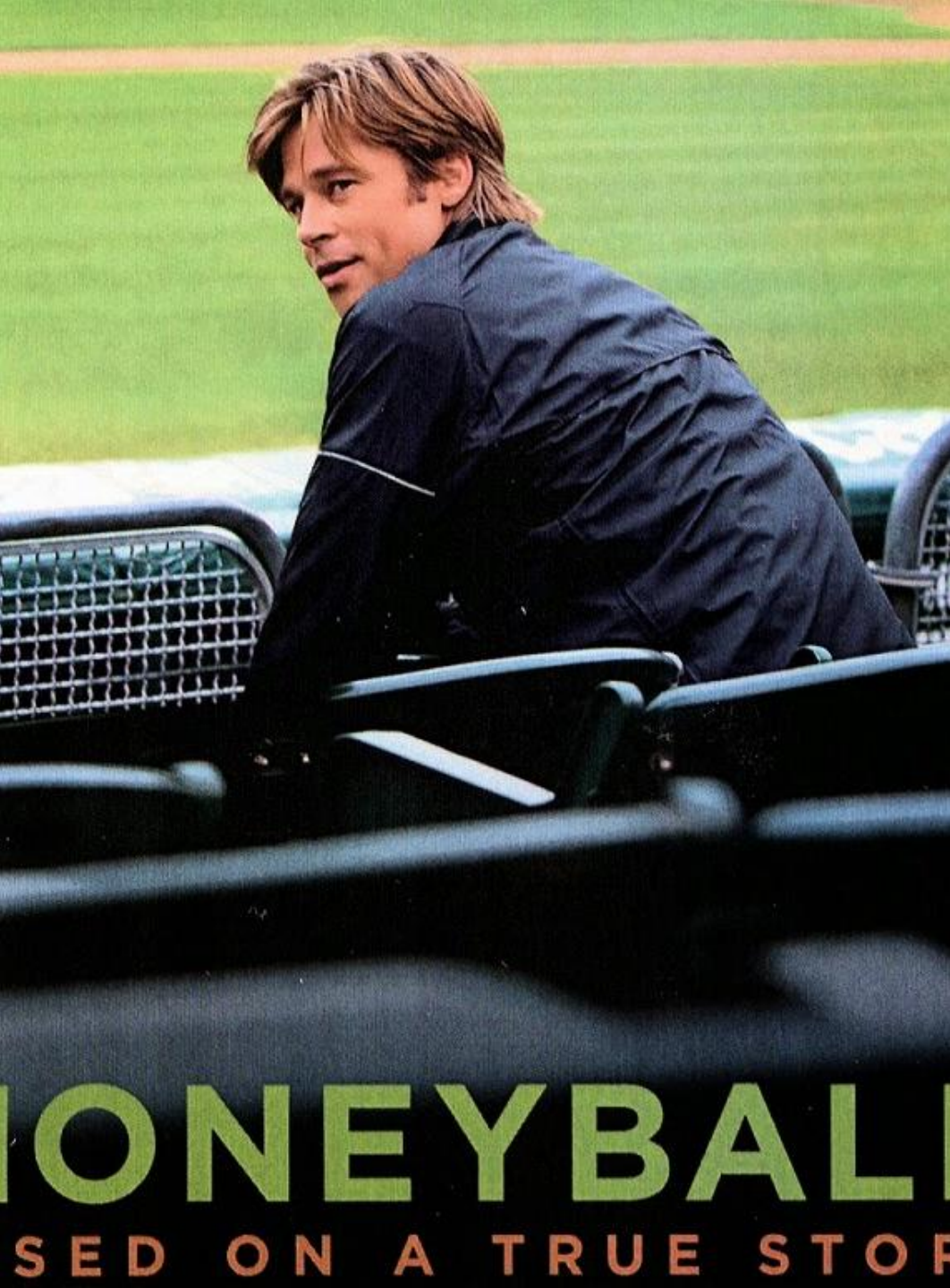
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Bill James quoted in *The Fielding Bible* (John Dewan): “In my career I have spent far more hours studying fielding than I have studying hitting or pitching. I just don’t have much to show for it.”

He proposed a statistic called *fielding range*, computed by the sum of putouts plus assists divided by games played



Here’s a comparison of Puckett, Hunter, Buxton, and the Say Hey kid Willie Mays.



EVOLUTION OF ADVANCED METRICS

Bill James and Sabermetrics

- Often considered the “father of advanced baseball analytics”
- Defined as “the search for objective knowledge about baseball”
- His analysis and writings about baseball data analytics started to get traction in the 1980’s
 - Introduced ideas such as WAR (Wins Above Replacement)
- Named to the list of 100 most influential people by Time in early 2000’s
- Hired by the Red Sox in 2003 and they subsequently won 2 World Series
- Concepts carried on the Society for American Baseball Research (SABR)



Moneyball: The Art of Winning an Unfair Game

- Book by Michael Lewis on General Manager Billy Beane's use of Data Analytics to transform the Oakland A's
- In 2002 the A's payroll was more than \$20M of rival teams
- Based upon his experience as a prospect for the NY Mets, Beane thought that current player scouting methods were inadequate
 - Instincts versus data
- Focused instead on Sabermetric approaches
 - On-base percentage + Slugging percentage to evaluate hitters
- Results
 - Made the playoffs in 2002, with record 20 game win streak
 - Made the playoffs 8 times during his tenure
 - Won only one playoff series (2006 vs Minnesota Twins)
 - Never played in league championship series or won a World Series
 - Approach adopted by many teams and now all teams have a "Data Analytics Group in the Front Office helping guide strategy and in-game decision making"